Annex - A
PATENT SEARCH REPORT

Title

Banana Peel Coagulant Formulation and the process of making thereof

Filing Date

Abstract (*Not more than 150 words*)

Industries are required to adhere to the water quality guidelines and general effluent standards set by the Philippine government. Coagulation, one of the important processes in wastewater treatment, is where dissolved and suspended solid particulates are aggregated by the use of an agglomerating agent (coagulant), increasing its size and mass, have them settle down, turning them to sludge for ease of removal form the system. This present utility model relates to production of coagulating agent for wastewater treatment but more particularly to an esterified banana peel as coagulating agent. The banana peels are dried and size reduced, then modified by the addition of methanol and an acid solution in a controlled environment to yield esterified banana peels, and then oven dried to produce the coagulant powder. This product allows the removal of suspended solids, oil and grease, and dissolved solids in wastewater.

International Patent Classification
C02F1/54; use of organic material
C02F1/56; macromolecular compounds
C08F 8/14; Esterification

Keywords
Esterification banana peels coagulating agent wastewater treatment organic solid waste

SEARCH STRINGS AND DATABASES

Database	Search String	Number of Hits
Espacenet	(organic solid waste) AND (coagula* or “coagula* agent”)	61 331
Espacenet	(C02F1/54) AND (coagula* or “coagula* agent”)	2 110
Espacenet	(C02F1/56) AND (coagula* or “coagula* agent”)	10 286
Espacenet	(organic solid waste) AND (banana AND (peel or skin)) AND (coagula* or “coagula* agent”)	646
Espacenet	(C02F1/54) AND (banana AND (peel or skin)) AND (coagula* or “coagula* agent”)	5
Espacenet	(C02F1/56) AND (banana AND (peel or skin)) AND (coagula* or “coagula* agent”)	11
Espacenet	(C02F1/56) AND (banana AND (peel or skin)) AND (coagula* or “coagula* agent”) AND ((“waste water*” OR wastewater) AND treat*)	10

Espacenet	(organic solid waste) AND (banana AND (peel or skin)) AND (coagulant or coagulating agent) AND (water treatment OR watertreatment)	107
Espacenet	(C02F1/56) AND (banana AND (peel or skin)) AND (ester*) AND (coagula* or "coagula* agent")	6
Espacenet	(organic solid waste) AND (banana AND (peel or skin)) AND (ester*) AND (coagula* or "coagula* agent")	536
Espacenet	(organic solid waste) AND (banana AND (peel or skin)) AND (ester*) AND (coagula* or "coagula* agent") AND (water treatment OR watertreatment)	523
Espacenet	(C02F1/56) AND (banana AND (peel or skin)) AND (ester*) AND (coagula* or "coagula* agent") AND (water treatment OR watertreatment)	6
Espacenet	(C02F1/56) AND (banana AND (peel or skin)) AND (ester*) AND (coagula* or "coagula* agent") AND (("waste water*" OR wastewater) AND treat*)	5
Espacenet	(C02F1/56) AND (banana AND (peel or skin)) AND (C08F 8/14) AND (coagula* or "coagula* agent") AND (("waste water*" OR wastewater) AND treat*)	0
	((banana AND (peel or skin)) AND (ester*) AND (coagula* or "coagula* agent") AND (("waste water*" OR wastewater) AND treat*)	78

DOCUMENTS CONSIDERED TO BE RELEVANT
(Citation of Documents, with indication, where appropriate, of the relevant pages)

Category	Document Information			Relevant to Claim No.
	Patent No.	Title	Publication Date	
A	CN105692850A	Preparation technique of banana peel polymer coagulant for water treatment	2016-06-22	
A	CN106830255A	Coagulation technology of banana skin water treatment agent and without mixing process	2017-06-13	
A	CN105883996A	Modified banana peel coagulant and preparation process thereof	2016-08-24	
A	CN106219715A	Preparation method of oxidation modification banana peel coagulant	2016-12-14	
A	CN114804330A	Applications method of pericarp coagulant and external carbon source for low-carbon-source sewage	2022-07-29	
A	WO2010131492A1	FLOCCULANT COMPOSITION	2010-11-18	
A	CN102583681A	Plant-sourced flocculant, flocculant mixture, condensation method and method for preparing plant-sourced flocculant	2012-07-18	
A	WO2013090569A2	TREATMENT OF WASTEWATER	2013-06-20	
A	US10562799B2	Systems and methods for removing finely dispersed particulate matter from a fluid stream	2020-02-18	
A	KR20230020444A	덱스트란-알파-글루칸 그래프트 공중합체 및 이의 유도체	2023-02-10	

*** Special Categories of Cited Documents:**

"A" document defining the general state of the art which is not considered to be of particular relevance

"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the

document is taken alone

"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

Note: Further references not indicated in this search report may be cited during substantive examination.

CERTIFICATION

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